Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{sr(IN)}$ $t_{sf(IN)}$	Data input setup time	-	3	-	-	ns
$t_{hr(IN)}$ $t_{hf(IN)}$	Data input hold time	-	1	-	-	
$t_{vr(OUT)}$ $t_{vf(OUT)}$	Data output valid time	DHQC = 0	-	3.5	4.5	
		DHQC=1 All prescaler values (except 0)	-	$t_{(CLK)}/4$ + 0.5	$t_{(CLK)}/4$ + 1.5	
$t_{hr(OUT)}$ $t_{hf(OUT)}$	Data output hold time	DHQC = 0	1.5	-	-	
		DHQC=1, All prescaler values (except 0)	$t_{(CLK)}/4$ - 0.5	-	-	

1. *DHQC must be set to reach the mentioned frequency.*

Table 56. XSPI characteristics in DTR mode (with DQS) / HyperBus

Evaluated by characterization - Not tested in production.
All values apply to Octal- and Quad-SPI mode.
Delay block activated.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$f_{(CLK)}$	Clock frequency	2.7 V < V_{DD} < 3.6 V C_L=15 pF	-	-	72[1][2]	MHz
$t_{w(CLKH)}$	OCTOSPI clock high and low time (even division)	PRESCALER[7:0] = n =(0,1,3,5,..,255)	$t_{(CLK)}/2$ - 0.5	-	$t_{(CLK)}/2$ + 0.5	ns
$t_{w(CLKL)}$			$t_{(CLK)}/2$ - 0.5	-	$t_{(CLK)}/$ 2 + 0.5	
$t_{w(CLKH)}$	OCTOSPI clock high and low time (odd division)	PRESCALER[7:0] = n =(2,4,6,..,254)	(n/2)×$t_{(CLK)}$/(n+1) - 0.5	-	(n/2) × $t_{(CLK)}$/(n+1) + 0.5	
$t_{w(CLKL)}$			(n/2+1)×$t_{(CLK)}$/(n+1) - 0.5	-	(n/2+1) × $t_{(CLK)}$/ (n+1) + 0.5	
$t_{v(CLK)}$	Clock valid time	-	-	-	$t_{(CLK)}$ + 2	
$t_{h(CLK)}$	Clock hold time	-	$t_{(CLK)}/2$ - 1	-	-	
$t_{w(CS)}$	Chip select high time	-	3 × $t_{(CLK)}$	-	-	ns
$t_{v(DQ)}$	Data input valid time	-	0	-	-	
$t_{v(DS)}$	Data strobe input valid time	-	0	-	-	
$t_{h(DS)}$	Data strobe input hold time	-	0	-	-	
$t_{v(RWDS)}$	Data strobe output valid time	-	-	-	3 × $t_{(CLK)}$	
$t_{sr(DQ)}$ $t_{sf(DQ)}$	Data input setup time	-	-1[3]	-	-	
$t_{hr(DQ)}$ $t_{hf(DQ)}$	Data input hold time	-	3	-	-	
$t_{vr(OUT)}$ $t_{vf(OUT)}$	Data output valid time	DHQC = 0	-	3.5	4.5	
		DHQC = 1, All prescaler values (except 0)	-	$t_{(CLK)}/4$ + 0.5	$t_{(CLK)}/$ 4 + 1.5	
$t_{hr(OUT)}$ $t_{hf(OUT)}$	Data output hold time	DHQC = 0	1.5	-	-	
		DHQC = 1, All prescaler values (except 0)	$t_{(CLK)}/4$ - 0.5	-	-	

1. *Maximum frequency values are given for a RWDS to DQ skew of maximum ±1.0 ns.*